

1 Curves in $\mathbb{R}^3$

1.1 Basic Definitions of curves

- **Length of a Curve:** The length of a parameterized curve c is $L(c) := \int_a^b \| \dot{c}(t) \| dt$.
- **Regular Curve:** A curve $c(t)$ is regular if $\dot{c}(t) \neq 0$ for all $t \in I$.
- **Arc Length Parameterization:** A curve $c(s)$ is unit-speed if $\| \dot{c}(s) \| = 1$.
For exerty regular curve, $\psi(t) = \int_{t_0}^t \| \dot{c}(t) \| dt$, $\psi^{-1} \equiv \varphi$, $\dot{c} = c \circ \varphi$ is unit-speed.

- **Frenet Frame:** If c is a unit-speed curve, $T(t) := \dot{c}(t)$, $N(t) := \frac{\ddot{c}(t)}{\| \ddot{c}(t) \|}$ has unit length nad are perpendicular to each other.

- **Curvature (κ):** Measures how fast the curve turns. For a unit-speed curve, $\kappa(t) = \frac{\| \dot{c}(t) \|}{\| \ddot{c}(t) \|}$.

Generally, $\kappa(t) = \frac{\| \dot{c}(t) \times \ddot{c}(t) \|}{\| \dot{c}(t) \|^3} = \frac{\sqrt{\| \ddot{c}(t) \|^2 - \dot{\varphi}(t)^2}}{v(t)^2} = \frac{\sqrt{v(t)^2 \| \ddot{c}(t) \|^2 - (\dot{c}(t) \cdot \ddot{c}(t))^2}}{v(t)^3}$

- **Torsion (τ):** Measures how much the curve twists out of the plane of curvature.
For a general curve, $\tau(t) = \frac{(\dot{c}(t) \times \ddot{c}(t)) \cdot \ddot{\dot{c}}(t)}{\| \dot{c}(t) \times \ddot{c}(t) \|^2}$.

- **Frenet-Serret Frame:** Let c be a unit-speed curve in $\mathbb{R}^3$, $T(t) = \dot{c}(t)$, $N(t) = \frac{\ddot{c}(t)}{\| \ddot{c}(t) \|} = \frac{\ddot{c}(t) - \kappa(t) T(t)}{\| \kappa(t) \|}$, $B(t) = T(t) \times N(t)$ are the tangent field, the normal field and the binormal field. $\{T, N, B\}$ is called the Frenet-Serret frame.

- **Frenet-Serret Formulas:** $\begin{pmatrix} T' \\ N' \\ B' \end{pmatrix} = \begin{pmatrix} 0 & \kappa & 0 \\ -\kappa & 0 & \tau \\ 0 & -\tau & 0 \end{pmatrix} \begin{pmatrix} T \\ N \\ B \end{pmatrix}$

1.2 Extra Applications

- **Isoperimetric Inequality:** For a closed curve of length L enclosing area A , $L^2 \geq 4\pi A$ with equality iff the curve is a circle.
- **Winding Number:** Let c a loop and P_0 be a point not lying on the curve, the winding number $W(c, P_0)$ is the number of times c goes counter clockwise around P_0 .
If $c(t) = P_0 + r(t) (\cos \theta(t), \sin \theta(t))$ and $P_0 = 0$, then $W(c, P_0) = \frac{1}{2\pi} \int_a^b \frac{\dot{y}x - \dot{x}y}{x^2 + y^2}(t) dt$.

1.3 Calculation Examples

For a plane curve $c(t) = (x(t), y(t))$, the curvature is given by: $\kappa = \frac{| \dot{x}\ddot{y} - \dot{y}\ddot{x} |}{(\dot{x}^2 + \dot{y}^2)^{3/2}}$.

2 Surfaces in $\mathbb{R}^3$

2.1 Definition and Construction of Surfaces

- **Regular Map:** A map Φ is called regular if $\frac{\partial \Phi}{\partial x}(p)$ and $\frac{\partial \Phi}{\partial y}(p)$ are not zero and not parallel.

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Then the first and second fundamental forms at m are given by $I^* = \begin{pmatrix} E & F \\ F & G \end{pmatrix}$, $II^* =$

$$\begin{pmatrix} e & f \\ f & g \end{pmatrix}, S = II \times I^{-1} = \frac{1}{EG - F^2} \begin{pmatrix} e & f \\ f & g \end{pmatrix} \begin{pmatrix} G & -F \\ -F & E \end{pmatrix}.$$

- **Area Element under Local Parametrization 2:** $dA = \sqrt{EG - F^2} du dv$.

2.8 Curvature

- **Normal Curvature:** The normal curvature in the u -direction is $\kappa_p(u) = II_p(u, u) = \langle u, S_p(u) \rangle$.

- **Principal Directions and Curvatures:** Principal curvature of S at p is the maximum and minimum values of $\kappa_p(u)$ for $\|u\| = 1$, $u \in T_pS$. The corresponding directions are called principal directions.

The principal curvatures and directions are the eigenvalues and eigenvectors of the shape operator S_p .

- **Gaussian Curvature:** $K = \kappa_1 \kappa_2$ is the determinant of the shape operator.
- **Mean Curvature:** $H = \frac{\kappa_1 + \kappa_2}{2}$.

- **Curvatures under Local Parametrization:** $K = \det(S) = \frac{eg - f^2}{EG - F^2}$, $H = \frac{1}{2} \text{tr}(S) = \frac{eG - 2fF + gE}{2(EG - F^2)}$. The normal curvatures in direction $w = a\Phi_u + b\Phi_v$ is given by $\kappa \left(\frac{w}{\|w\|} \right) = \frac{ea^2 + 2Fab + gb^2}{Ea^2 + 2Fab + Gb^2}$. Use K and H to find principal curvatures.

- **Umbilic Point:** A point $p \in S$ is called an umbilic point if $\kappa_1(p) = \kappa_2(p)$
Otherwise, $\kappa_1(p)\kappa_2(p) > 0$ is called elliptic point, $\kappa_1(p)\kappa_2(p) < 0$ is called hyperbolic point, and $\kappa_1(p)\kappa_2(p) = 0$ is called parabolic point.

- **Darboux Frame:** An adapted frame on $\mathbb{R}^3$ is called a Darboux frame if $e_1(p), e_2(p)$ are the principal directions of S at p .

- **A Line of Curvature:** A unit-speed curve is called a line of curvature if $\dot{c}(t)$ is always a principal direction of S at $c(t)$. **Cal Method:** Φ is the parametrization of S , then a regular curve $c(t) = \Phi(x(t), y(t))$ is a line of curvature iff $(fE - eF)\dot{x}^2 + (gE - eG)\dot{y}^2 + (gF - fG)\dot{y}^2 = 0$.

2.9 Some equations of frames and curvatures

For an adapted frame e_1, e_2, e_3 and its connection forms and dual frames, we have

- $dw^1 = \omega_1^1 \wedge w^1 + \omega_2^1 \wedge w^2 = \omega_2^1 \wedge w^1$
- $\omega_1^1 \wedge w^1 + \omega_2^1 \wedge w^2 = 0$
- $\omega_2^1 \wedge w^1 + \omega_1^2 \wedge w^2 = 2Hw^1 \wedge w^2$
- $d\omega_1^2 = -Kw^1 \wedge w^2$, $d\omega_2^1 = \omega_1^1 \wedge \omega_2^1 + \omega_2^1 \wedge \omega_2^2 = \omega_2^1 \wedge \omega_1^3$
- $\omega_1^3 = II_{11}w^1 + II_{12}w^2$, $\omega_2^3 = II_{21}w^1 + II_{22}w^2$

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Given a subset $S \subseteq \mathbb{R}^3$, a coordinate chart Φ around a point $m \in S$, consists of

- an open subset $D \subseteq \mathbb{R}^2$
- a regular, injective smooth map $\Phi : D \rightarrow S$, $\Phi(D) \subseteq S$
- an open subset $\theta \in \mathbb{R}^3$, $m \in \theta$

such that $\Phi(D) = S \cap \theta$ and $\Phi^{-1} : \theta \cap S \rightarrow D$ is continuous.

- **Surface:** A subset $S \subseteq \mathbb{R}^3$ is a surface if every point $m \in S$ has a coordinate chart centered at m .

There are three common surfaces:

- **Ruled Surface:** A ruled surface is a surface swept only by straight line $l : I \rightarrow \mathbb{R}^3$ moving along some curve $c : I \rightarrow \mathbb{R}^3 \rightarrow \Phi(t, r) = c(t) + r\ell(t)$, $r \in \mathbb{R}$
- **Graphs:** Evert smooth function of $D \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}^3$ determines a parameterized surface $S = \text{graph } f = \{(x, y, z) \in D \times \mathbb{R} : z = f(x, y)\}$
- **Level Sets:** Let $g : \mathbb{R}^3 \rightarrow \mathbb{R}$ a smooth function. For every $c \in \mathbb{R}$, consider $S = g^{-1}(C) = \{(x, y, z) \in \mathbb{R}^3 : g(x, y, z) = c\}$ is called a level set.
If $\left(\frac{\partial g}{\partial x} \frac{\partial g}{\partial y} \frac{\partial g}{\partial z} \right) \neq 0$ on $g^{-1}(C)$, then S is a surface.
- **Smooth Map:** Given a surface S , a function $f : S \rightarrow \mathbb{R}^3$ is called smooth if all compositions $f \circ \Phi : \mathbb{R}^2 \supset D \rightarrow \mathbb{R}^3$ are smooth for every coordinate chart $\Phi : D \rightarrow S$.

2.2 Vector Fields and Tangent Space

- **Tangent Vector:** A tangent vector X at $m \in S$ is $X \in \mathbb{R}^3$ such that there exists a smooth curve $c : (-\epsilon, \epsilon) \rightarrow S$ with $c(0) = m$ and $\dot{c}(0) = X$.
- **Tangent Plane:** The set of all tangent vectors at $m \in S$ is called the tangent plane at m , denoted by T_mS .

Cal Method: Take the derivative of the surface ∇F , and the tangent plane is $\{w \in \mathbb{R}^3 : \nabla F \cdot w = 0\}$

- **Vector Field:** A vectord field on S is a map $X : S \rightarrow \mathbb{R}^3$ such that $X(m) \in T_mS$ for all $m \in S$.

- **Directional Derivative:** Directional detivative of $f : S \rightarrow \mathbb{R}^k$ in the direction $X \in T_mS$ is defined by $T_m f(X) = \frac{d}{dt} f(m + tX)$.

- **Vector field under local parametrization:** Given a coordinate chart $\Phi(u, v) : D \rightarrow S$, a tangent vector X on S is the span of $\partial_u \Phi, \partial_v \Phi$.
The tangent plane at $m = \Phi(u_0, v_0)$ is $T_m \Phi(x, y) = x\partial_u^* \Phi(m) + y\partial_v^* \Phi(m)$, $(x, y) \in \mathbb{R}^2$.

2.3 Differential Forms

- **1-Form on Surface:** A 1-form on S is a linear map $\alpha_m : T_mS \rightarrow \mathbb{R}$ for each point $m \in S$. We write α for the whole family. $\alpha_m(v) = \alpha_m(v_1 \partial x + v_2 \partial y) = v_1 \alpha_m(\partial x) + v_2 \alpha_m(\partial y)$.

For Darboux frame, we have

$$d\kappa_1(e_2) = (\kappa_1 - \kappa_2)\omega_1^1(e_1), d\kappa_2(e_1) = (\kappa_1 - \kappa_2)\omega_1^2(e_2)$$

2.10 Gauss Curvature and Theorema Egregium

- **Push Forward and Full back:** Let $F : \mathbb{R}^2 \supseteq S \rightarrow \overline{S} \subseteq \mathbb{R}^3$
 - **Push Forward of Curves:** For a curve $c : I \rightarrow S$, the push forward of c by F is $F_*c = F \circ c : I \rightarrow \overline{S}$.
 - **Push Forward of Vectors:** For a vector $v \in T_pS$, $c(t) : I \rightarrow S$ s.t. $\dot{c}(0) = v$, the push forward of X by F is $F_*v = (F \circ c)'(0) = T_p F(\dot{c}(0))$, it's just the derivative $T_p F : T_pS \rightarrow T_{F(p)}\overline{S}$.
 - **Pull Back of functions:** For a function $f : \overline{S} \rightarrow \mathbb{R}$, the pull back of f by F is $F^*f = f \circ F : S \rightarrow \mathbb{R}$.
 - **Pull Back of 1-Forms:** For a 1-form α on $\overline{S}$, the pull back of α by F is $(F^*\alpha)_p(v) = \alpha_{F(p)}(T_p F(v))$ for all $p \in S$, $v \in T_pS$.
 - **Pull Back of 2-Forms:** The pull back of 2-form β by F is $(F^*\beta)_p(u, v) = \beta_{F(p)}(T_p F(u), T_p F(v))$ for all $p \in S$, $u, v \in T_pS$.
 - **Propositions:** If α, β are 1-forms on $\overline{S}$, then $F^*(\alpha \wedge \beta) = (F^*\alpha) \wedge (F^*\beta)$, $F^*(da) = d(F^*\alpha)$.

- **Isometry:** A bijective map $F : S \rightarrow \overline{S}$ is called an isometry if F preserves the inner product of tangent vectors $\langle v, w \rangle = \langle T_p F(v), T_p F(w) \rangle$, $\forall v, w \in T_pS$, i.e. $F^*I = I$.

- **Local Isometry:** A map $F : S \rightarrow \overline{S}$ is called a local isometry if for every $p \in S$, there exists an open neighborhood $O \subseteq S$ of p such that the restriction $F|_O : O \rightarrow F(O)$ is an isometry.

- **Distance on Surface:** For $p, q \in S$, the distance between p and q on S is defined by $d(p, q) = \inf \{L(c)\}$, $c(0) = p$, $c(1) = q$.

- **Isometry preserves distance:** If $F : S \rightarrow \overline{S}$ is an isometry, then $d(p, q) = \tilde{d}(F(p), F(q))$ for all $p, q \in S$.

- **Theorema Egregium:** The Gauss curvature K only depends on the first fundamental form I on S and it's invariant under local isometries $F : S \rightarrow \overline{S}$, i.e. $K(p) = \tilde{K}(F(p))$ for all $p \in S$.

- **Behaviour of frame under isometry:** If $F : S \rightarrow \overline{S}$ is an isometry, and e_1, e_2, e_3 is an adapted frame on S , then $\tilde{e}_1 = F_*e_1 = T_p F(e_1(p))$ is an adapted frame on $\overline{S}$. Dual frame $\{\omega^1, \omega^2\}$ is uniquely determined by $F^*\omega^i = \omega^i$ for $i = 1, 2$. ω_1^2 is uniquely determined by $F^*\omega_1^2 = \omega_1^2$.

- **Surface with all points umbilic:** Let S be a connected surface such that all points are umbilic, S has constant Gaussian curvature $K \geq 0$. And if $K = 0$, then S is a part of plane. If $K > 0$, then S is a part of sphere.

- **Constant Gaussian Curvature Surface:** Let S be a compact surface in $\mathbb{R}^3$ with constant

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1-form α is called closed if $d\alpha = 0$ and is called exact if there exists a function $f : S \rightarrow \mathbb{R}$ such that $\alpha = df$. exact implies closed.

- **1-Form** $(dx)_p, (dy)_p$: 1-form $(dx)_p$ is defined by $(dx)_p(\partial x)_p = 1$, $(dx)_p(\partial y)_p = 0$ and similarly for $(dy)_p$. Moreover, every 1-form α at p can be expressed as $\alpha = \alpha_1(p)(dx)_p + \alpha_2(p)(dy)_p$, α_1, α_2 are smooth functions.
- $T_p f$: Given a smooth map $f : S \rightarrow \mathbb{R}$, the directional derivative $T_p f$ is a 1-form at p defined by $\langle df \rangle_p(v) = T_p f(v) = (\nabla f) \cdot v$.

- **Local Representation of 1-Form:** Given a coordinate chart $\Phi(u, v) : D \rightarrow S$, any 1-form α on D , then $\alpha_{\Phi(x, y)} = \alpha_{\Phi(x, y)} \circ T_{(x, y)} \Phi$.
Cal Method: If $\alpha = \alpha_1 dx + \alpha_2 dy$, then $\alpha^* = \alpha_1(\Phi(u, v)) dx^* + \alpha_2(\Phi(u, v)) dy^*$.
- **Integration of 1-Form along Curve:** Given a smooth curve $c : [a, b] \rightarrow S$ and a 1-form α on S , the integral of α along c is defined by $\int_c \alpha := \int_a^b \alpha_{c(t)}(\dot{c}(t)) dt$.

Speciallt, if $\alpha = df$ and f is a smooth function, then $\int_c df = f(c(b)) - f(c(a))$.

- **2-Form:** A 2-form β on S is, for each point $m \in S$, β , a bilinear antisymmetric map $\beta_m : T_mS \times T_mS \rightarrow \mathbb{R}$. In particular, $\beta_m(v, v) = 0$.
- **k-Form:** A k-form ω on S is, for each point $m \in S$, a multilinear antisymmetric map $\omega_m : (T_mS)^k \rightarrow \mathbb{R}$.
- **Wedge Product:** Given two 1-form α_1, α_2 , their wedge product $\alpha_1 \wedge \alpha_2$ is a 2-form defined by $(\alpha_1 \wedge \alpha_2)_m(u, v) = (\alpha_1)_m(u)(\alpha_2)_m(v) - (\alpha_1)_m(v)(\alpha_2)_m(u)$, $(u, v) \in T_mS$.

Note that $\alpha_1 \wedge \alpha_2 = -\alpha_2 \wedge \alpha_1$.

- **Exterior Derivative of 1-Form:** α is a 1-form on S , its exterior derivative da is a 2-form defined by $(da)_m(\partial_u^*(m), \partial_v^*(m)) = T_m(\alpha(\partial_u^*(m))(\partial_v^*(m)) - T_m(\alpha(\partial_v^*(m))(\partial_u^*(m)))$ for a chart $\Phi : D \rightarrow S$.
Cal Method: If $\alpha = \alpha_1 du + \alpha_2 dv$, then $da = \left(\frac{\partial \alpha_2}{\partial u} - \frac{\partial \alpha_1}{\partial v} \right) du \wedge dv$.

- **Properties of Exterior Derivative:** $f : S \rightarrow \mathbb{R}^3$, α 1-form, $d(f\alpha) = df \wedge \alpha + f da$, $d(df) = 0$

2.4 Cartan's Moving Frame

- **Moving Frame:** A Moving frame on an open subset $U \subseteq \mathbb{R}^3$ consists of vector fields $e_1, e_2, e_3 : U \rightarrow \mathbb{R}^3$ such that for every $x \in U$, $e_1(x), e_2(x), e_3(x)$ are an orthonormal basis of $\mathbb{R}^3$.
- **Connection 1-Form:** Given a moving frame e_1, e_2, e_3 on S , the connection 1-form ω_i^j is defined by $(de_i)_x(v) = \sum_{j=1}^3 (\omega_i^j)_x(v) e_j(x)$. In short, $de_i = \sum_{j=1}^3 \omega_i^j e_j$. For ONB, $(\omega_i^j)_x(v) = \langle de_i \rangle_x(v) \cdot e_j(x)$.
- **Connection Form and Frenet-Serret Frame:** For a unit-speed curve c with $\kappa > 0$. Suppose e_1, e_2, e_3 is an moving frame on $\mathbb{R}^3$ such that the restrictions of the fields to c give the Frenet-Serret frame T, N, B . Then $(\omega_1^2)_{c(t)}(\dot{c}(t)) = \kappa$, $(\omega_1^3)_{c(t)}(\dot{c}(t)) = 0$, $(\omega_2^3)_{c(t)}(\dot{c}(t)) = \tau$.

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Gaussian curvature $K > 0$, then S is a sphere of radius $\frac{1}{\sqrt{K}}$.

2.11 Geodesics

- **Straight Line in Plane:** A curve $c(t)$ is a straight line $\iff \ddot{c}(t) = 0 \iff \dot{c}(t)$ are parallel $\iff c$ is the shortest path between two points.
- **Geodesic Curvature:** Let $\{e_1, e_2, e_3\}$ be an adapted frame such that $e_1(c(t)) = \frac{\dot{c}(t)}{v}$, i.e. e_1 is the unit tangent vector of c , e_3 is the unit normal vector of S , $e_2 = e_1 \times e_3$, then the geodesic curvature of c at $c(t)$ is defined by $\kappa_g(t) = \omega_1^2 \left(\frac{\dot{c}(t)}{v} \right) = \omega_1^2(e_1) = \langle e_1, e_2 \rangle$.
- **Other Cal Method:** If n is the normal vector of S , $r(t)$ is the parametrization of c , then $\kappa_g(t) = \frac{\det(\dot{r}(t), \ddot{r}(t), n(r(t)))}{\| \dot{r}(t) \|^3} = \frac{(\dot{r}(t) \times \ddot{r}(t)) \cdot n(r(t))}{\| \dot{r}(t) \|^3}$.
- **The relation between curvatures:** Let $\kappa(t)$ be the curvature of c at $c(t)$ in $\mathbb{R}^3$, and $\kappa_n(t)$ be the normal curvature of S at $c(t)$ in the direction of $\dot{c}(t)$, then $\kappa^2(t) = \kappa_g^2(t) + \kappa_n^2(t)$ and $\kappa_n(t) = \omega_1^3 \left(\frac{\dot{c}(t)}{v} \right) = \omega_1^3(e_1) = \frac{e_1^T \cdot n}{\| \dot{c}(t) \|}$.
- **Geodesic:** A curve c on S is called a geodesic $\iff \kappa_g(t) = 0 \iff \ddot{c}(t)$ is in the normal direction $\iff \omega_1^2(\dot{c}(t)) = 0 \iff c$ is locally the shortest path between two points on S .
- **Cal Method:** If $\Phi(u, v)$ is the parametrization of S , then a regular curve $c(t) = \Phi(x(t), y(t))$ is a geodesic iff $\ddot{x} + \frac{1}{2E}(E_x \dot{x}^2 + 2E_x \dot{x}\dot{y} - G_x \dot{y}^2) = 0$ and $\ddot{y} + \frac{1}{2G}(G_x \dot{x}^2 + 2G_x \dot{x}\dot{y} - E_y \dot{y}^2) = 0$.

- **Vector Field along Curve:** A vector field X along a curve $c : I \rightarrow S$ is a map $X : I \rightarrow \mathbb{R}^3$ such that $X(t) \in T_{c(t)}S$ for all $t \in I$.

- **Covariant Derivative:** Let e_1, e_2, e_3 be a frame and $X(t) = f_1(t)e_1(t) + f_2(t)e_2(t)$ be a vector field along c . The covariant derivative of X along c is the vector field $\frac{\nabla}{dt} X$ along c defined by $\frac{\nabla}{dt} X(t) = \left(\dot{f}_1 + f_2 \omega_1^2(c) \right) e_1(c(t)) + \left(\dot{f}_2 + f_1 \omega_1^2(c) \right) e_2(c(t))$ for all $t \in I$.

- **Gauss-Weingarten Formula:** Let e_1, e_2, e_3 be an adapted frame on S , and $X(t)$ be a vector field along a curve $c : I \rightarrow S$. Then the derivative of X in $\mathbb{R}^3$ can be decomposed as $\frac{d}{dt} X(t) = \frac{\nabla}{dt} X(t) + II_{c(t)}(\dot{c}(t), X(t)) \cdot e_3(c(t))$.

- **Covariant Derivative and Geodesic:** Let c be a unit-speed curve and $\{e_1, e_2, e_3\}$ an adapted frame with $e_1(c(t)) = \dot{c}(t)$, then $\frac{\nabla}{dt}(\dot{c}(t)) = \kappa_g(t)e_2(c(t))$, $\kappa_n(t) = \kappa(\dot{c}(t))$. Moreover, c is a geodesic $\iff \frac{\nabla}{dt}(\dot{c}(t)) = 0$.

- **Parallel Transport:** Given a curve $c : I \rightarrow S$ and a vector $X_0 \in T_{c(t_0)}S$, there exists a unique vector field X along c such that $X(0) = X_0$ and $\frac{\nabla}{dt} X(t) = 0$ for all $t \in I$. Such X is called the parallel transport along c from X_0 along c .
- **Holonomy Angle:** Given a curve $c : I \rightarrow S$ and the tangent frame $\{e_1, e_2\}$, the holonomy angle of the parallel transport along c is defined by $\text{Hol}(c) = - \int_c \omega_1^2$.

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- **Dual Frame:** Dual frames are 1-forms ω^i defined by $\omega^i(e_j) = \delta_j^i$.

Cal Method: Given a chart $\Phi(u, v, w)$, then $\begin{cases} dx = \sum_{i=1}^3 \omega^i e_i \\ dx = \frac{\partial \Phi}{\partial u} du + \frac{\partial \Phi}{\partial v} dv + \frac{\partial \Phi}{\partial w} dw \end{cases}$, so we

- can solve ω^i .
- **First Cartan's Structure Equation:** $d\omega^k + \sum_i \omega_i^k \wedge \omega^i = 0$.
- **Second Cartan's Structure Equation:** $d\omega_i^j + \sum_k \omega_k^j \wedge \omega_i^k = 0$.

2.5 Adapted Frame

- **Adapted Frame:** A moving frame $e_1, e_2, e_3 : O \rightarrow \mathbb{R}^3$ is called adapted to the surface S if for every $p \in S \cap O$, $e_3(p)$ is normal to S , i.e. $e_3(p) \times X = 0$ for all $X \in T_pS$. In particular, if S is oriented and n is a unit normal field on S , then $n(p) = \pm e_3(p)$ for all $p \in S \cap O$. $e_1(p), e_2(p)$ are a basis of $T_p(S)$

- $\kappa = \frac{-(r(t),0) \cdot (-r)}{r^2} = 0$.
- **Mean Curvature:** $H = -1/(2r)$.
 - **Geodesics:** Assume $\gamma(t) = x(t), y(t)$, then $r^2\ddot{x} = 0, \ddot{y} = 0 \Rightarrow x(t) = at + b, y(t) = ct + d$.

3.1.2 Example: Torus

- Parametrization: $\Phi(u, v) = ((R + r \cos v) \cos u, (R + r \cos v) \sin u, r \sin v)$.
- **Gaussian Curvature:** $K = \frac{r(R + r \cos v)}{r^2}$
 - Outer circle ($v = 0$): $K > 0$. Inner circle ($v = \pi$): $K < 0$. Top/Bottom ($v = \pm\pi/2$): $K = 0$.

3.1.3 Example: Surface of Revolution

Curve $c(t) = (r(t), 0, z(t))$ rotated around z-axis.

- **Gaussian Curvature:** $K = -\frac{r''(t)}{r(t)}$ (assuming unit speed profile)

3.2 Algorithm: Computing Curvature via Coframe

- Choose a Frame:** Find orthonormal e_1, e_2 tangent to S .
- Find Coframe:** Write $ds^2 = (\omega^1)^2 + (\omega^2)^2$. Identify ω^1, ω^2 .
- Differentiate:** Compute $d\omega^1$ and $d\omega^2$.
- Solve for ω_{12} :** Use the First Structure Equation:

$$d\omega^1 = \omega^2 \wedge \omega_{21}, \quad d\omega^2 = \omega^1 \wedge \omega_{12}$$

- (Recall $\omega_{21} = -\omega_{12}$).
- **Compute K :** Calculate $d\omega_{12}$ and write it as $-K\omega^1 \wedge \omega^2$.

3.3 The Darboux Curve (Frame along a Curve)

- For a unit-speed curve $c(s)$ on a surface S :
- **Definition:** An orthonormal frame $\{T, V, n\}$ adapted to both the curve and the surface.
 - $T(s) = \dot{c}(s)$ (Unit Tangent).
 - $n(s)$ is the surface unit normal restricted to the curve.
 - $S(s) = n(s) \times T(s)$ (Tangent Normal / Side Normal).

- **Darboux Equations** $\begin{pmatrix} T' \\ S' \\ n' \end{pmatrix} = \begin{pmatrix} 0 & -\kappa_g & -\kappa_n \\ \kappa_g & 0 & -\tau_g \\ \kappa_n & \tau_g & 0 \end{pmatrix} \begin{pmatrix} T \\ S \\ n \end{pmatrix}$

- **Curvatures:**
 - **Geodesic Curvature** (κ_g): Bending in the tangent plane. $\kappa_g = \langle T', V \rangle = \omega_{12}(T)$.
 - **Normal Curvature** (κ_n): Bending normal to the surface. $\kappa_n = \langle T', n \rangle = II(T, T) = \omega_{13}(T)$.
 - **Geodesic Torsion** (τ_g): Twisting of the normal. $\tau_g = \langle n', V \rangle = \omega_{23}(T)$.
- **Relation to Space Curvature:** The curvature vector of the curve is $\vec{\kappa} = T' = \kappa_g V + \kappa_n n$. Squared curvature: $\kappa^2 = \kappa_g^2 + \kappa_n^2$.

$$\begin{aligned} & \text{Take the integral, } \int \frac{cy}{\sqrt{1-c^2y^2}} dy = \frac{-1}{2c} \int \frac{1}{\sqrt{1-c^2y^2}} d(1-c^2y^2) = -\frac{1}{2c} \int \frac{1}{u^{\frac{1}{2}}} du = \\ & -\frac{1}{c} \sqrt{1-c^2y^2}. \end{aligned}$$

$$\text{Therefore, we have } -\frac{1}{c} \sqrt{1-c^2y^2} = \pm(x-x_0) \Rightarrow (x-x_0)^2 + y^2 = \left(\frac{1}{c}\right)^2$$

4.5 EXERCISE 5.

$$\text{Curve } c(t) = \left(t, \frac{1+t}{t}, \frac{1-t^2}{t}\right) \text{ for } t \in (0, \infty).$$

- (i) **Prove that c is regular.**
 $\dot{c}(t) = (1, -t^{-2}, -t^{-2} - 1)$. The first component is always 1, thus $\|\dot{c}(t)\| \neq 0$ for all t .
- (ii) **Compute the torsion of c .**

$$\dot{c}(t) = (0, 2t^{-3}, 2t^{-3}). \text{Then } \dot{c} \times \dot{c} = \begin{vmatrix} i & j & k \\ 1 & -t^{-2} & -1-t^{-2} \\ 0 & 2t^{-3} & 2t^{-3} \end{vmatrix} = 2t^{-3}(-1, -1, 1).$$

$$\text{Therefore, the curvature } \kappa \text{ is given by: } \kappa = \frac{\|\dot{c} \times \dot{c}\|}{\|\dot{c}\|^3} = \frac{2\sqrt{3}/t^3}{\left(\frac{\sqrt{2(t^4+t^2+1)}}{t^2}\right)^3} = \frac{\sqrt{6}t^3}{(t^4+t^2+1)^{3/2}}$$

- (iii) **Compute the torsion of c .**

$$\begin{aligned} \ddot{c} &= (0, -6t^{-4}, -6t^{-4}). (\dot{c} \times \ddot{c}) \cdot \ddot{c} = 2t^{-7}(-1, -1, 1) \cdot (-6t^{-4}(0, 1, 1)) = -12t^{-7}(-1+1) = 0. \\ \text{Thus, the torsion } \tau &= \frac{(\dot{c} \times \ddot{c}) \cdot \ddot{c}}{\|\dot{c} \times \ddot{c}\|^2} = \frac{0}{\|2t^{-3}(-1, -1, 1)\|^2} = 0. \end{aligned}$$

4.6 EXERCISE 6.

Let S be a compact surface with unit normal field $n : S \rightarrow S^2$.

- (i) **Prove that n is surjective onto S^2 .**
 Let $v \in S^2$ be an arbitrary unit vector in $\mathbb{R}^3$. We want to show that there exists a point $p \in S$ such that $n(p) = v$.
 Consider the *height function* $h_v : S \rightarrow \mathbb{R}$ defined by the dot product: $h_v(x) = x \cdot v$ for all $x \in S$.
 Since S is compact and h_v is continuous, h_v must achieve its global maximum at some point $p \in S$.
 Let p be a point where h_v attains its maximum. At a critical point of a function defined on a surface, the gradient of the restriction is zero. Geometrically, this means the tangent plane $T_p S$ is orthogonal to the gradient of the ambient linear function $f(x) = x \cdot v$. The gradient of f is simply the vector v . Therefore, the vector v is normal to the tangent plane $T_p S$.
 This implies that the unit normal $n(p)$ is parallel to v . That is: $n(p) = v$ or $n(p) = -v$.
 To distinguish between these two, observe that p is a global maximum. This means that for all $x \in S, x \cdot v \leq p \cdot v$. Geometrically, the entire surface S lies in the half-space defined by the plane passing through p with normal v .

4.2 Test Exam

4.1 EXERCISE 1. (True/False)

Are the following claims true or false? Give a short one sentence explanation for your answer.

- (i) For any given smooth functions $\kappa > 0$ and τ (defined on some interval) there exists a regular curve with curvature κ and torsion τ .
Answer: True. Curvature and torsion uniquely determine a curve.
- (ii) The subset of $\mathbb{R}^3$ defined by the equation $z = x^2 - y^2$ is a surface.
Answer: True. It can be seen as level set $g(x, y, z) = x^2 - y^2 - z = 0$. $\nabla g = (2x, -2y, -1) \neq 0$.
- (iii) Every surface admits a non-zero three-form.
Answer: False. If $n > m$, then a n -form on an m -dimensional manifold is always zero. Since a surface is 2-dimensional, any 3-form on it must be zero.
- (iv) The principal directions of a surface are intrinsic.
Answer: False. The principal direction is the eigenvector of the shape operator, and the definition of the shape operator depends on the choice of the normal vector n in the ambient space and the embedding, so the principal direction is an extrinsic geometric quantity.
- (v) If two isometric, connected surfaces in $\mathbb{R}^3$ have the same second fundamental form, then one is the image of the other under a composition of a translation, a rotation and a reflection.
Answer: True. The Fundamental Theorem of Surfaces gets this.
- (vi) If v is a tangent vector at a point p and S_p the shape operator, then $S_p(v)$ is a vector normal to the surface.
Answer: False.
- (vii) A curve c in $\mathbb{R}^3$ of constant speed is a geodesic if and only if the acceleration is tangent to the surface S at $c(t)$ for all t .
Answer: False. On the contrary, the definition of a geodesic is that the tangent vector field is parallel transported along the curve, which means the acceleration vector $\ddot{c}(t)$ has no tangent component and must be **normal** to the surface (i.e., $\ddot{c}(t) \parallel n$).

4.2 EXERCISE 2.

Let θ, φ be the standard spherical coordinates on S^2 , and consider the function $f : S^2 \rightarrow \mathbb{R}$ given by $f(x, y, z) = (x - y)^2 + z^2$. Calculate df in terms of θ, φ .

The parameterization of a unit sphere is given by $\Phi(\theta, \varphi) = (\sin \theta \cos \varphi, \sin \theta \sin \varphi, \cos \theta)$.

Method 1: Then we have $f = (x - y)^2 + z^2 = x^2 - 2xy + y^2 + z^2 = (x^2 + y^2 + z^2) - 2xy = 1 - 2(\sin \theta \cos \varphi)(\sin \theta \sin \varphi) = 1 - 2\sin^2 \theta \sin \varphi \cos \varphi = 1 - \sin^2 \theta \sin(2\varphi)$

Take the derivative, we get $df = \frac{\partial f}{\partial \theta} d\theta + \frac{\partial f}{\partial \varphi} d\varphi = (-2 \sin \theta \cos \theta \sin(2\varphi)) d\theta + (-\sin^2 \theta \cdot 2 \cos(2\varphi)) d\varphi = -\sin(2\theta) \sin(2\varphi) d\theta - 2\sin^2 \theta \cos(2\varphi) d\varphi$

$$\begin{aligned} & \text{Assuming the standard convention that } n \text{ is the } \textit{outward} \text{ pointing normal of a compact surface, the normal at the "highest" point in the direction } v \text{ must point in the direction of } v \text{ (out of the bounded region enclosed by } S). \text{ Therefore, } n(p) = v. \\ \text{Since } v \in S^2 \text{ was arbitrary, for every unit vector } v, \text{ there exists a point } p \in S \text{ such that } n(p) = v. \text{ Thus, } n \text{ is surjective.} \end{aligned}$$

- (ii) **Let $K^+(p) = \max\{0, K(p)\}$. Show that $\int_S K^+ dA \geq 4\pi$.**
 Since n is surjective, for every unit vector $v \in S^2$, there exists at least one point $p \in S$ such that $n(p) = v$. Each such point p corresponds to a global maximum of the height function in the direction of v , and at these points, the Gaussian curvature $K(p)$ is non-negative. By considering the area of the sphere and the contributions from these points, we can show that the total positive curvature must be at least 4π .
- (iii) **Determine the image of n for the catenoid and the cylinder.**
 For the Catenoid, the normal vector is $n = (\text{sech } v \cos u, \text{sech } v \sin u, -\tanh v)$. It covers the entire sphere except the north and south poles as v varies over $(-\infty, \infty)$.
 For the Cylinder, the normal vector is $n = (\cos u, \sin u, 0)$. It only covers the equator of the sphere as u varies over $[0, 2\pi)$.

4.7 EXERCISE 7.

- Compact, oriented surface S with mean curvature H and Gauss curvature K .
- (i) **Prove that $H^2 \geq K$.**
 By the definition, $H = \frac{\kappa_1 + \kappa_2}{2}, K = \kappa_1 \kappa_2, H^2 - K = \left(\frac{\kappa_1 + \kappa_2}{2}\right)^2 - \kappa_1 \kappa_2 = \frac{(\kappa_1 - \kappa_2)^2}{2} \geq 0$.
 - (ii) **Show that S cannot be a minimal surface.**
 If S were a minimal surface, then $H = 0$ for all points on S . From (i), we have $0 = H^2 \geq K \Rightarrow K \leq 0$ for all points on S .
 But S is a compact surface, there exists at least one point p such that $K(p) > 0$. This contradicts the previous result that $K \leq 0$ everywhere on S .
 Therefore, S cannot be a minimal surface.
 - (iii) **Assume S is not the sphere and that is does not have a smooth bijection to a sphere. Prove that then there are points on S where the Gaussian curvature is positive, negative and zero.**
 Since S is compact, there exists at least one point p such that $K(p) > 0$.
 By the Gauss-Bonnet theorem, $\int_S K dA = 2\pi \chi(S)$. Since S is not a sphere, its Euler characteristic $\chi(S) \leq 0$. Thus, the total Gaussian curvature $\int_S K dA \leq 0$. Thus, there must be points on S where $K < 0$ to ensure that the total curvature is non-positive.
 Since S is connected and K is a continuous function, by the Intermediate Value Theorem, there must be points on S where $K = 0$ as well.

Method 2: Alternatively, we can compute df directly using the chain rule.

$$\begin{aligned} df &= \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy + \frac{\partial f}{\partial z} dz = 2(x - y)dx + 2(y - x)dy + 2zdz \\ &= 2(\sin \theta \cos \varphi - \sin \theta \sin \varphi)(\cos \theta \cos \varphi d\theta - \sin \theta \sin \varphi d\varphi) \\ &\quad + 2(\sin \theta \sin \varphi - \sin \theta \cos \varphi)(\cos \theta \sin \varphi d\theta + \sin \theta \cos \varphi d\varphi) + 2(\cos \theta)(-\sin \theta d\theta) \\ &= 2\sin \theta (\cos \varphi - \sin \varphi)(\cos \theta \cos \varphi - \sin \varphi) d\theta - 2\cos \theta \sin \theta \sin \varphi d\varphi \\ &= -\sin(2\theta) \sin(2\varphi) d\theta - 2\sin^2 \theta \cos(2\varphi) d\varphi \end{aligned}$$

4.3 EXERCISE 3.

- The catenoid S is given by $\Psi(u, v) = (\cosh v \cos u, \cosh v \sin u, v), u \in [0, 2\pi), v \in \mathbb{R}$.
- (i) **Determine an adapted moving frame $\{e_1, e_2, e_3\}$ on S .**
 Take partial derivatives: $\Psi_u = (-\cosh v \sin u, \cosh v \cos u, 0), \Psi_v = (\sinh v \cos u, \sinh v \sin u, 1)$. These two vectors are orthogonal, thus we can utilize them to get the tangent frame e_1, e_2 .

$$e_1 = \frac{\Psi_u}{\|\Psi_u\|} = \frac{\Psi_u}{\cosh v} = (-\sin u, \cos u, 0), e_2 = \frac{\Psi_v}{\|\Psi_v\|} = \frac{\Psi_v}{\sqrt{\sinh^2 v + 1}} = \frac{\Psi_v}{\cosh v} = (\tanh v \cos u, \tanh v \sin u, \frac{1}{\cosh v})$$

$$e_3 = e_1 \times e_2; e_3 = \begin{vmatrix} i & j & k \\ -\sin u & \cos u & 0 \\ \tanh v \cos u & \tanh v \sin u & \text{sech } v \end{vmatrix} = (\text{sech } v \cos u, \text{sech } v \sin u, -\tanh v)$$
 - (ii) **Calculate the dual frame $\{\omega^1, \omega^2\}$ and the connection forms ω^i_j .**
 Since $e_1 = \frac{\partial u}{\cosh v}$. And $\omega^1(e_1) = 1$, thus $\omega^1 = \cosh v du$. The same reason to the $\omega^2 = \cosh v dv$.
 Another solution is $\begin{cases} dx = \omega^1 e_1 + \omega^2 e_2 \\ dx = \Phi_u du + \Phi_v dv = \cosh v e_1 du + \cosh v e_2 dv \end{cases}$ and we can get the same result.

$$\omega^2 = de_1 e_2 = (-\cos u du, -\sin u du) \cdot (\tanh v \cos u, \tanh v \sin u, \text{sech } v) = -\tanh v \cos^2 u du - \tanh v \sin^2 u du = -\tanh v du.$$

 The same reason to get $\omega^1_2 = -\text{sech } v dv, \omega^2_2 = -\text{sech } v dv$.
 - (iii) **Calculate the mean curvature H and the Gaussian curvature K of S .**
 $d\omega^1_2 = -K\omega^1 \wedge \omega^2$ and $d\omega^2_1 = d(-\tanh v du) = -\text{sech}^2 v dv \wedge du = \text{sech}^2 v du \wedge dv, \omega^1 \wedge \omega^2 = \cosh^2 v du \wedge dv$.
 Compare the coefficients, we get $K = -\frac{1}{\cosh^4 v}$.
 For mean curvature H , we have $\omega_{13} \wedge \omega^1 + \omega^1 \wedge \omega_{23} = 2H\omega^1 \wedge \omega^2$.
 Substitute the values, we get $H = 0$.
 - (iv) **Verify that the total Gaussian curvature $\int_S K dA$ of S is -4π** ($\frac{d}{dv} \tanh v = \frac{1}{\cosh^2 v}$).
 $dA = \omega^1 \wedge \omega^2 = \cosh^2 v du \wedge dv$.
 Thus we have $\int_S K dA = \int_{-\infty}^{\infty} \int_0^{2\pi} \left(-\frac{1}{\cosh^4 v}\right) (\cosh^2 v) du dv$

$$= \int_{-\infty}^{\infty} \int_0^{2\pi} -\frac{1}{\cosh^2 v} du dv = 2\pi \int_{-\infty}^{\infty} -\text{sech}^2 v dv$$

$$= -2\pi [\tanh v]_{-\infty}^{\infty} = -2\pi(1 - (-1)) = -4\pi$$

$$\begin{aligned} & \text{The same reason to get } \omega^1_2 = -\text{sech } v dv, \omega^2_2 = -\text{sech } v dv. \\ & \text{Calculate the mean curvature } H \text{ and the Gaussian curvature } K \text{ of } S. \\ & d\omega^1_2 = -K\omega^1 \wedge \omega^2 \text{ and } d\omega^2_1 = d(-\tanh v du) = -\text{sech}^2 v dv \wedge du = \text{sech}^2 v du \wedge dv, \omega^1 \wedge \omega^2 = \cosh^2 v du \wedge dv. \\ & \text{Compare the coefficients, we get } K = -\frac{1}{\cosh^4 v}. \\ & \text{For mean curvature } H, \text{ we have } \omega_{13} \wedge \omega^1 + \omega^1 \wedge \omega_{23} = 2H\omega^1 \wedge \omega^2. \\ & \text{Substitute the values, we get } H = 0. \end{aligned}$$

$$\begin{aligned} & \text{(iv) Verify that the total Gaussian curvature } \int_S K dA \text{ of } S \text{ is } -4\pi \left(\frac{d}{dv} \tanh v = \frac{1}{\cosh^2 v} \right). \\ & dA = \omega^1 \wedge \omega^2 = \cosh^2 v du \wedge dv. \\ & \text{Thus we have } \int_S K dA = \int_{-\infty}^{\infty} \int_0^{2\pi} \left(-\frac{1}{\cosh^4 v} \right) (\cosh^2 v) du dv \\ & = \int_{-\infty}^{\infty} \int_0^{2\pi} -\frac{1}{\cosh^2 v} du dv = 2\pi \int_{-\infty}^{\infty} -\text{sech}^2 v dv \\ & = -2\pi [\tanh v]_{-\infty}^{\infty} = -2\pi(1 - (-1)) = -4\pi \end{aligned}$$

5 Appendix

5.1 Integrals of Special Functions

+C and dx is omitted in the following formulas for brevity.

$$\begin{aligned} & \text{Trigonometric Functions} \\ & \int \tan x = -\ln |\cos x| \\ & \int \cot x = \ln |\sin x| \\ & \int \sec x = \ln |\sec x + \tan x| \\ & \int \csc x = \ln |\csc x - \cot x| \\ & \text{Inverse Trigonometric Functions} \\ & \int \arcsin x = x \arcsin x + \sqrt{1-x^2} \\ & \int \arccos x = x \arccos x - \sqrt{1-x^2} \\ & \int \arctan x = x \arctan x - \frac{1}{2} \ln(1+x^2) \\ & \int \text{arccot } x = x \text{arccot } x + \frac{1}{2} \ln(1+x^2) \\ & \int \text{arcsec } x = x \text{arcsec } x - \ln|x + \sqrt{x^2-1}| \\ & \int \text{arccsc } x = x \text{arccsc } x + \ln|x + \sqrt{x^2-1}| \\ & \text{Hyperbolic Functions} \\ & \int \sinh x = \cosh x \\ & \int \cosh x = \sinh x \\ & \int \tanh x = \ln(\cosh x) \\ & \int \coth x = \ln|\sinh x| \\ & \int \text{sech } x = \arctan(\sinh x) \\ & \int \text{csch } x = \ln \left| \tanh \frac{x}{2} \right| \\ & \text{Inverse Hyperbolic Functions} \\ & \int \text{arsinh } x = x \text{arsinh } x - \sqrt{x^2+1} \\ & \int \text{arcosh } x = x \text{arcosh } x - \sqrt{x^2-1} \\ & \int \text{artanh } x = x \text{artanh } x + \frac{1}{2} \ln(1-x^2) \\ & \int \text{arcoth } x = x \text{arcoth } x + \frac{1}{2} \ln(x^2-1) \\ & \int \text{arsech } x = x \text{arsech } x + \arcsin x \\ & \int \text{arsch } x = x \text{arsch } x + \text{arsinh } x \\ & \text{Rational/Irrational Integrals} \\ & \int \frac{1}{\sqrt{x^2+1}} = \ln|x + \sqrt{x^2+1}| \\ & \int \frac{1}{\sqrt{x^2-1}} = \ln|x + \sqrt{x^2-1}| \\ & \int \frac{1}{\sqrt{1-x^2}} = \arcsin x \\ & \int \frac{1}{\sqrt{a^2-x^2}} = \frac{1}{|a|} \arctan \left| \frac{a}{x} \right| \\ & \int \frac{1}{x^2-a^2} = \frac{1}{2|a|} \ln \left| \frac{x-a}{x+a} \right| \\ & \int \frac{1}{\sqrt{x^2+a^2}} = \frac{x}{2\sqrt{x^2+a^2}} + \frac{a^2}{2} \ln|x + \sqrt{x^2+a^2}| \\ & \int \frac{1}{\sqrt{x^2-a^2}} = \frac{x}{2\sqrt{x^2-a^2}} - \frac{a^2}{2} \ln|x + \sqrt{x^2-a^2}| \\ & \int \frac{1}{\sqrt{a^2-x^2}} = \frac{x}{2\sqrt{a^2-x^2}} + \frac{a^2}{2} \arcsin \frac{x}{|a|} \\ & \int \frac{1}{\sqrt{(x^2+a^2)^3}} = \frac{x}{a^2\sqrt{x^2+a^2}} \\ & \int \frac{1}{\sqrt{(x^2-a^2)^3}} = -\frac{x}{a^2\sqrt{x^2-a^2}} \end{aligned}$$

5.2 Hyperbolic Trigonometric Identities

$$\begin{aligned} & \sinh x = \frac{e^x - e^{-x}}{2}, \cosh x = \frac{e^x + e^{-x}}{2} \\ & \tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}} \\ & \cosh^2 x - \sinh^2 x = 1 \end{aligned}$$

5.3 EXERCISE 4.

- Assume the first fundamental form od a surface S in a certain parametrization is given by $I_{ij} = \begin{pmatrix} 1/y^2 & 0 \\ 0 & 1/y^2 \end{pmatrix}$ on the half-plane $D = \{y > 0\}$.
- (i) **Calculate a tangent frame field e_1, e_2 on D .**
 Using the first fundamental form, we get $\langle \partial_x, \partial_x \rangle = \frac{1}{y^2}$, thus $\|\partial_x\| = \frac{1}{y}$, so we have $e_1 = y\partial_x$. Similarly, we get $e_2 = y\partial_y$.
 - (ii) **Determine the connection form ω^1_2 uniquely defined by the structure equation.**
 Firstly, we can get the dual frame $\omega^1 = \frac{1}{y} dx, \omega^2 = \frac{1}{y} dy$.
 Assume $\omega^2 = Adu + Bdv$, then we have $d\omega^1 = \omega^1_2 \wedge \omega^2$ and $d\omega^2 = \omega^1 \wedge \omega^2_1$.

$$d\omega^1 = d\left(\frac{1}{y} dx\right) = -\frac{1}{y^2} dy \wedge dx = \frac{1}{y^2} dx \wedge dy = \frac{1}{y^2} (A dx + B dy) \wedge \frac{1}{y} dy = \frac{A}{y} dx \wedge dy \Rightarrow A = \frac{1}{y}$$

$$d\omega^2 = -\frac{1}{y^2} dy \wedge dy = 0 = \left(\frac{1}{y}\right) dx \wedge (A dx + B dy) = \frac{B}{y} dx \wedge dy \Rightarrow B = 0$$

 Thus we have $\omega^1_2 = \frac{1}{y} dx$.
 - (iii) **Determine the geodesics.**

The first fundamental form is $I_{ij} = \begin{pmatrix} E & F \\ F & G \end{pmatrix} = \begin{pmatrix} 1/y^2 & 0 \\ 0 & 1/y^2 \end{pmatrix}$. We know that a regular curve $\gamma(t) = (x(t), y(t))$ on the surface is a geodesic if it satisfies the following system of differential equations: $\ddot{x} + \frac{1}{2E}(E_x \dot{x}^2 + 2E_y \dot{x}\dot{y} - G_x \dot{y}^2) = 0, \ddot{y} + \frac{1}{2G}(G_y \dot{y}^2 + 2G_x \dot{x}\dot{y} - E_g \dot{x}^2) = 0$.
Given $E = y^{-2}, G = y^{-2}$, and $F = 0$, the partial derivatives are: $E_x = \frac{\partial}{\partial x}(y^{-2}) = 0, E_y = \frac{\partial}{\partial y}(y^{-2}) = -2y^{-3}, G_x = \frac{\partial}{\partial x}(y^{-2}) = 0, G_y = \frac{\partial}{\partial y}(y^{-2}) = -2y^{-3}$.
Substituting into the first equation, we have $\ddot{x} + \frac{y}{2} ((-2y^{-3})\dot{x}^2 + 2(0)\dot{x}\dot{y} - (-2y^{-3})\dot{y}^2) = \ddot{x} - \frac{2\dot{y}\dot{y}}{y} = 0$.
And the same to the second equation, $\ddot{y} + \frac{y}{2} ((-2y^{-3})\dot{y}^2 + 2(0)\dot{x}\dot{y} - (-2y^{-3})\dot{x}^2) = \ddot{y} - \frac{1}{y}(\dot{x}^2 - \dot{y}^2) = 0$.

Case 1: $\dot{x} = 0$, then $x(t) = x_0, \dot{x} = 0$, therefore $\dot{y} + \frac{1}{y}\dot{y}^2 = 0 \Rightarrow y = \sqrt{At + B}$.

Case 2: $\dot{x} \neq 0$, from the first equation, we have $\frac{\ddot{x}}{\dot{x}} = \frac{2\dot{y}}{y}$, integrating both sides with respect to t , we get $\ln|\dot{x}| = 2\ln|y| + C_1 \Rightarrow \dot{x} = cy^2$.
Since the geodesic has constant speed, we can assume unit speed without loss of generality. The length element squared is $ds^2 = \frac{dx^2 + dy^2}{y^2} = 1$. Thus we have $\frac{\dot{x}^2 + \dot{y}^2}{y^2} = 1 \Rightarrow \dot{x}^2 + \dot{y}^2 = y^2$.
Substitute $\dot{x} = cy^2$ into the speed equation, we have $(cy^2)^2 + \dot{y}^2 = y^2 \Rightarrow c^2 y^4 + \dot{y}^2 = y^2 \Rightarrow \dot{y}^2 = y^2(1 - c^2 y^2)$.
Thus $\frac{dy}{dx} = y^2(1 - c^2 y^2) \Rightarrow \pm dx = \frac{cy}{\sqrt{1 - c^2 y^2}} dy$.

$$\begin{aligned} & \text{sech}^2 x + \tanh^2 x = 1 \\ & \coth^2 x - \text{csch}^2 x = 1 \\ & \sinh(x \pm y) = \sinh x \cosh y \pm \cosh x \sinh y \\ & \cosh(x \pm y) = \cosh x \cosh y \pm \sinh x \sinh y \\ & \tanh(x \pm y) = \frac{\tanh x \pm \tanh y}{1 \pm \tanh x \tanh y} \\ & \text{arcsinh } x = \ln x + \sqrt{x^2+1} \\ & \text{arcosh } x = \ln x + \sqrt{x^2-1} \\ & \text{artanh } x = \frac{1}{2} \ln \frac{1+x}{1-x} \end{aligned}$$

5.3 Translations of common Curves and Surfaces

- Cylinder: 圆柱面
- Torus: 环面
- Catenoid: 双曲面 (悬链面)
- Helicoid: 螺旋面